Tanut Bumrungvongsiri

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Summary

Artificial Intelligence engineer with 2+ years of experience building OCR and vision-language model pipelines for technical engineering drawings. Fine-tunes and deploys models for information extraction, metadata standardization, and computer vision applications, with project experience in reinforcement learning for games and robot navigation.

Education

King Mongkut's University of Technology Thonburi July 2020 – May 2024

Bachelor's in Robotics and Automation Engineering, Institute of Field Robotics *Bangkok, Thailand*

- GPAX: 3.74 / 4.00 (First-Class Honors)
- **Relevant Coursework:** Artificial Intelligence, Machine Learning, Pattern Recognition, Natural Language Processing, Reinforcement Learning, Image Processing, Calculus, Statistics & Probability

Experience

Artificial Intelligence / Machine Learning Engineer July 2024 – Present

Vulcorn AI · Full-time *Bangkok, Thailand (Remote)*

Digitalization for 2D Technical Engineering Drawings

- Designed and implemented an end-to-end pipeline for extracting textual and semantic information from 2D technical engineering drawings.
- Applied fine-tuned transformer-based OCR models to achieve high-accuracy text detection and recognition across complex drawing layouts.
- Leveraged fine-tuned Vision-Language Models (VLMs) to semantically understand and extract key metadata from 2D technical engineering drawings.

Digital Rebranding and Standardization for 2D Technical Drawings

- Implemented an automated pipeline to mask sensitive title block metadata in 2D engineering drawings and replace it with standardized, company-branded layouts.
- Utilized AI-based OCR and vision models to extract essential metadata while ensuring sensitive and proprietary information is protected.

Artificial Intelligence / Machine Learning Engineer & Software Developer June 2023 – November 2023

The Bangkok Unitrade Co., Ltd. · 6-month Internship *Bangkok, Thailand*

- Designed and adapted object detection for surgical instruments moving along a conveyor belt for classification, employing YOLOv8.
- Conducted research and applied a multi-view classification model utilizing data from three camera perspectives to fine-grained classify surgical instruments, achieving 99.4% accuracy on 44 surgical instrument classes.
- Developed a software application with a PyQt5 graphical user interface (GUI) integrating AI-powered object detection and classification.
- Implemented logging functionality to record predictions in a MySQL database.

Technical Skills

Programming Languages: Python (Advanced); C, C++ (Intermediate)

Frameworks: PyTorch, TensorFlow, Keras, OpenCV, Transformers (Hugging Face), Ultralytics, scikit-learn, LangChain, LangGraph, vLLM

Backend & Deployment: FastAPI, Docker, Linux, Neo4j (graph database)

Applied AI & ML: Model fine-tuning (including VLM-LoRA), end-to-end deployment, Retrieval-Augmented Generation (RAG), GraphRAG

Computer Vision: Object detection and feature recognition (YOLO, R-CNN); Image generation (GANs, Diffusion Models, VAEs); OCR pipelines for structured information extraction; Vision-Language Models (VLMs)

Natural Language Processing: Large Language Models (LLMs); multimodal text and image understanding

Reinforcement Learning: Q-Learning, Deep Q-Learning (DQN), Proximal Policy Optimization (PPO); continuous control (DDPG, TD3)

Projects

Santorini Board Game with a Self-Trained PPO Agent October 2026

- Implemented the PPO-Clip algorithm from scratch and trained policy and value networks through self-play across 8 processes and 16 boards.
- Trained a residual CNN agent on 2.0M games. The Hard agent achieved 98.8% against the tactical baseline and 100% against random and one-move lookahead baselines, evaluated over 400 games per opponent with equal games from each seat.
- Exported the PyTorch weights to run the agent in the browser using a Web Worker, deployed as a static site on GitHub Pages.

Note: the web demo contains AI-generated code.

Vehicle Tracking using YOLOv8 + SORT June 2024

- Leveraged YOLOv8 for vehicle detection and implemented tracking with SORT and Kalman filters for accurate estimation and tracking.

Deep Learning-based Navigation for Mobile Robots in Dynamic Environments May 2024

Team project — my contribution: adapting TD3 and SAC for the deep reinforcement learning navigation component.

- Adapted TD3 and SAC deep reinforcement learning algorithms for mobile robot navigation in dynamic environments with moving obstacles in Gazebo simulation.
- Grouped LiDAR clusters and tracked obstacles in each frame using the Hungarian Algorithm and Kalman Filter.

Solving Tic-Tac-Toe with Reinforcement Learning April 2024

- Trained an agent to play Tic-Tac-Toe using Q-Learning, SARSA, and Double Q-Learning, and compared the performance of the three algorithms.
- Extended with a Deep Q-Network for a 4×4 variant and an interactive browser demo that shows each agent's Q-values.

Image Classification of Surgical Instruments March 2023

- Compared performance of fine-tuned ResNet-50 and VGG-16 models pre-trained on ImageNet, using data augmentation.
- Integrated the Attention Network mechanism with SEResNet50, enhancing the model's capability to focus on relevant features.
- Applied fine-grained classification methods (Bilinear Pooling, WSDAN); reported project accuracy of 98.5% across 4 surgical instrument classes.

Telemanipulation of Robot Hand using Human Gesture November 2022

- Adapted MediaPipe and OpenCV to track 4 human gestures for controlling a universal robot arm and gripper.
- Implemented socket-based client–server communication to enable telemanipulation over the network.

Solving the 8-Puzzle and Mazes with AI October 2022

- Developed Pygame software utilizing the A* algorithm to solve the 8-puzzle with the minimum number of moves, and adapted A* search to maze pathfinding problems.

Publication

Automated Bounding Dimension Retrieval: A Machine Learning-Driven Framework for Engineering Drawing Interpretation - ICCI 2025 (Co-author)

Certification

Deep Learning for Image Classification — learn.thairobotics June 2024

You Only Look Once (YOLO): Unified, Real-Time Object Detection — learn.thairobotics June 2024

Image Processing — learn.thairobotics November 2022